

RippleTech controllers

REV manual

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Revision History

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v0.6.0	6/7/25	RC/BB	Bespoke manuals for REV/REP/REM.
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v0.5.2	23/5/24	RC/BB	
v0.5.1	20/2/24	RC/BB	
v0.5	8/11/23	RC/BB	

Contact Details

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Warranty

International

RippleTech Ltd Consumer Guarantee & Manufacturer's Warranty

Your Rights Under the Consumer Guarantees Act 1993

You are entitled to a replacement or refund for a major failure and for compensation for any other reasonably foreseeable loss or damage. You are also entitled to have the goods repaired or replaced if the goods fail to be of acceptable quality and the failure does not amount to a major failure.

RippleTech Ltd 12-Month Manufacturer's Warranty

In addition to your rights under the Consumer Guarantees Act, RippleTech Ltd warrants this product to be free from defects in materials and workmanship for a period of one (1) year from the date of original purchase.

- **What we will do:** If a defect arises during the warranty period, RippleTech will, at its option, (a) repair the product at no charge, (b) replace the product with a new or refurbished product, or (c) refund the original purchase price.
- **Exclusions:** This warranty does not apply to damage caused by improper installation, improper configuration, accident, abuse, misuse, modification, or other external causes.
- **How to claim:** To make a claim, please contact us at [email/phone] with your proof of purchase.

This manufacturer's warranty is provided by RippleTech Ltd, 385 State Highway 33, Rotorua 3074.



New Zealand

RippleTech Ltd Limited Warranty

1. Limited Warranty RippleTech Ltd warrants that this product will be free from defects in materials and workmanship for a period of one (1) year from the date of purchase ("Warranty Period"). This warranty applies only to the original commercial purchaser.

2. Exclusive Remedy If the product fails to conform to this warranty during the Warranty Period, RippleTech Ltd will, at its sole discretion, either repair or replace the defective product. This is the purchaser's sole and exclusive remedy.

3. Exclusions and Limitations This warranty is void if the product failure has resulted from: a) Installation, application, or maintenance not in accordance with RippleTech Ltd's instructions. b) Accident, neglect, misuse, modification, or unauthorized repair. c) Operation outside of the product's specified environmental or electrical parameters.

4. Disclaimer of Other Warranties EXCEPT FOR THE LIMITED WARRANTY EXPRESSLY STATED ABOVE, RIPPLETECH LTD PROVIDES ITS PRODUCTS "AS IS" AND DISCLAIMS ALL OTHER WARRANTIES, WHETHER EXPRESS, IMPLIED, OR STATUTORY, INCLUDING THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

5. Limitation of Liability IN NO EVENT SHALL RIPPLETECH LTD BE LIABLE FOR ANY INDIRECT, CONSEQUENTIAL, INCIDENTAL, SPECIAL, OR PUNITIVE DAMAGES (INCLUDING BUT NOT LIMITED TO LOSS OF PROFITS, LOSS OF USE, OR DAMAGE TO OTHER PROPERTY) ARISING FROM OR RELATING TO THE PRODUCT. RIPPLETECH LTD'S TOTAL CUMULATIVE LIABILITY SHALL IN NO EVENT EXCEED THE PURCHASE PRICE PAID FOR THE SPECIFIC PRODUCT GIVING RISE TO THE CLAIM.

6. Governing Law This Limited Warranty shall be governed by the laws of New Zealand, without regard to its conflict of law principles.



Notices

Notice To Operators This entire manual should be read before using the controller.

Unsafe Operation Using the SPUD controller to control a machine or vehicle without a complete and thorough understanding of the capabilities and limitations of the controller is unsafe and may cause harm. It is important that this manual be read and understood in its entirety before using the controller. Adherence to the statutory regulations regarding accident prevention, occupational safety and environmental protection is also important.

Warnings and Cautions Section Please read the section on Warnings and Cautions carefully before operating the controller.

Use and Maintenance Any questions regarding installing, operating, or maintaining the controller should be directed to RippleTech Ltd.



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1 INTRODUCTION

1.1 Introduction

This User Manual contains detailed information and instructions which when adhered to ensure the safe and effective set up, use and maintenance of the REV controller. The user should read the **entire** manual before using REV. REV is based on RippleTech's SPUD controller, however this manual contains everything required to connect and use REV in an EV conversion.

REV is a controller and firmware combination that provides a VCU for the following electric drive trains:

- Gen 1 Nissan Leaf (ZE0)
- Gen II Nissan Leaf (AZE0/ZE1)
- BYD Atto3¹
- Tesla Model 3/Y²

This manual describes electrical connection of REV to the vehicle, and how to modify the available firmware settings.

1.2 WARNINGS

Untrained Personnel Only properly trained personnel should operate the controller.

1.2.1 Fire or Explosion

Operation of the SPUD controller in the presence of flammable gases could cause a fire or explosion. Under no circumstances is the controller to be operated when explosive gases are present.

1.2.2 Maximum ratings

Operation where SPUD is subjected to more than the maximum ratings in the SPUD datasheet may damage the controller and will void any warranty.

¹This version of the controller is still in testing

²This version of the controller is still in testing



1.3 Cautions

1.3.1 ESD and EMI

Electrostatic discharge (ESD) and electromagnetic interference (EMI) can negatively affect the performance of SPUD.



2 CONNECTING

The front panel of REV has a single connector with the following connections:

- Power (x2)
- GND
- 5V output
- 6 analog inputs
- 4 digital inputs/outputs
- 6 high power switches
- Two CAN bus interfaces



Figure 1: Connecting REV to the machine is done via a single connector (IP40 version).

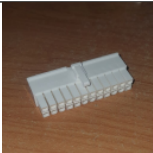


2.1 Main connector

The normal main connector is a 24-pin Minitex® Pwr 4.2 connector from Amphenol. A matching harness can be purchased separately or bundled with REV.

2.1.1 24-pin Minitex Pwr 4.2 connector

This connector is a male 24-pin Minitex® Pwr 4.2 Connector from Amphenol. The female wire connector and crimps can be purchased from RippleTech Ltd, as can ready-made pigtailed. Alternatively these parts can be purchased using the information below:

Item	Part number	Example
MINITEK MALE CONNECTOR HOUSING	10127815-24LF	
CONN SOCKET 16-22AWG CRIMP TIN	10127817-221LF	
CONN SOCKET 24-28AWG CRIMP TIN	10127817-222LF	
TOOL HAND CRIMPER 16-30AWG SIDE	10129483-001LF	

Note that alternative crimp tools are available.



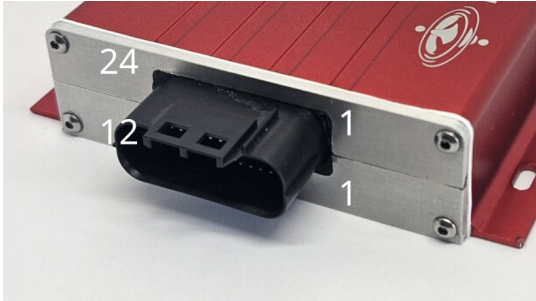
AI2	AI1	AI0	GND	CAN0-H	CAN1-H	SW5	SW4	SW3	SW2	SW1	SW0
24	23	22	21	20	19	18	17	16	15	14	13
12	11	10	9	8	7	6	5	4	3	2	1
AI3	AI4	AI5	5V	CAN0-L	CAN1-L	DIO0	DIO1	DIO2	DIO3	VCC2	VCC1

2.1.2 26-pin MX23A connector



Figure 2: IP64 variant of REV.

The REV controller can be purchased in an IP64 variant. The connector is a male 26-pin MX23A series connector from JAE. The female wire connector and crimps can be purchased from RippleTech Ltd, as can ready-made pigtails. Alternatively the customer can purchase these parts using the information below:



AI2	AI1	AI0	GND	CAN0-H	CAN1-H	SW5	SW4	SW3	SW2	SW1	SW0	VCC1
24											13	1
12											2	1
AI3	AI4	AI5	5V	CAN0-L	CAN1-L	DIO0	DIO1	DIO2	DIO3	VCC2	VCC2	VCC1



Item	Part number	Example
CONNECTOR HOUSING	10127815-24LF	
Front cap	MX23A26XF1	
Crimp	M23S05K351	
Dummy pin	M120-55780	

3 SETTING JUMPERS

3.1 CAN interfaces

There are two jumpers on each CAN interface. When the jumpers are in place the CAN bus is terminated at this interface. This should only be done if the CAN bus interface is at the end of the physical CAN network. This figure below gives an example of CAN1 being used at the end of the CAN network.



Figure 3: CAN1 as the terminal node on the CAN bus.



4 SETTING PARAMETERS

REV keeps certain settings between restarts of the vehicle. These can be modified using Wi-Fi. The procedure is as follows:

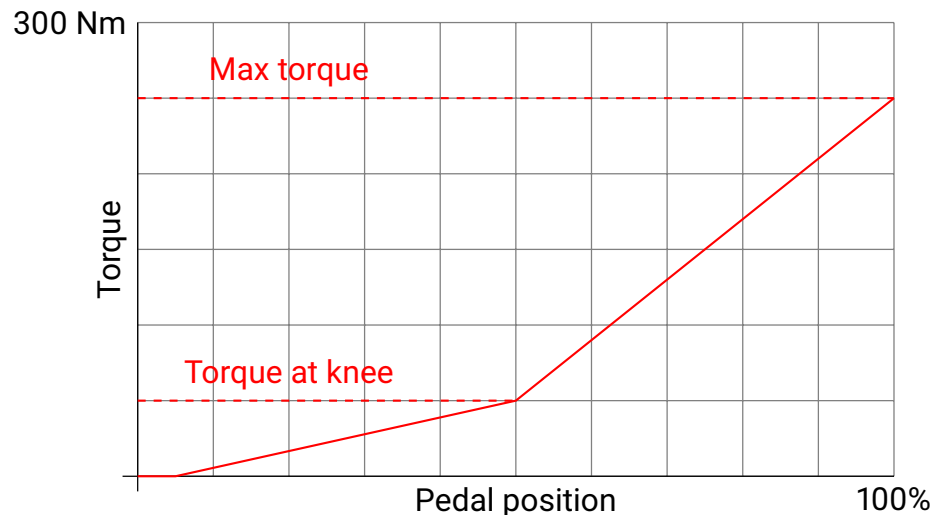
- Turn off the vehicle, but leave the isolation switch ON
- The REV's SSID should become available (default is SPUD)
- Enter the password (default password = spud1234)
- Use a browser on the same device and navigate to <http://192.168.1.1>

Access to these settings are only available after the power is applied (not ignition) to the controller and before the traction battery relays are engaged. This is by design so that WiFi is not available during operation.

The more complicated parameters are discussed below. All other parameters are explained using the "info" pop up in the settings page itself. If you see a setting that would be helpful, please let us know.

4.1 Torque at knee

The pedal map is controlled by two user settings: **max torque** and **torque at knee**. This is shown below.

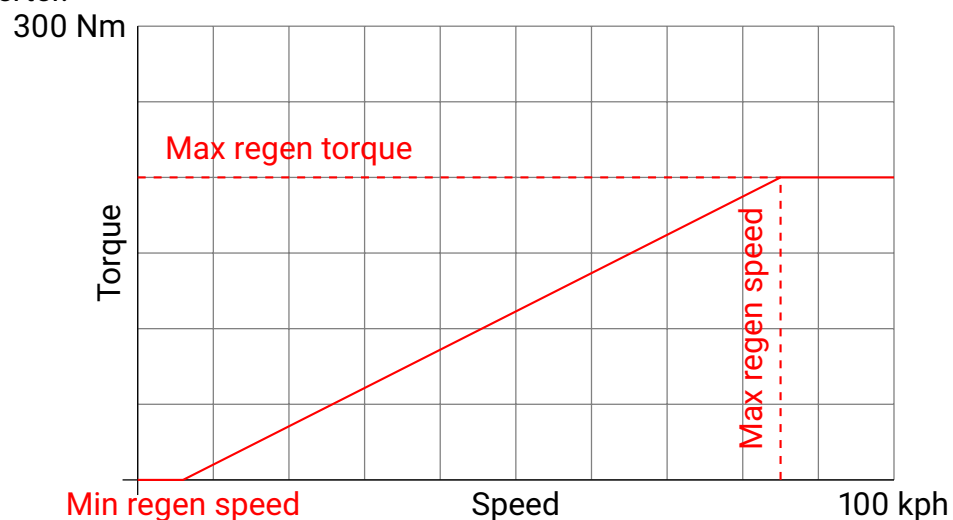




4.2 Regeneration

The regenerative braking is calculated from three settings as shown below. There is no regenerative braking at speeds below **min regen speed**, while the **max regen torque** is applied at speeds above **max regen speed**. Between these speeds a linear map is applied.

The inverter torque is the combination of these two maps. This means that high values of **max regen torque** will limit the high speed torque applied to the inverter.



5 DASH

RippleTech offers a 4.3" touchscreen dash with REV controllers. This can be bundled with REV or purchased separately. It allows the operator to control functions and view values, warnings and errors.



Figure 4: 4.3" dash with software



6 APPENDIX

6.1 Connecting REV to a drive train (general)



Figure 5: Connector J1 (VCU)

A three way switch is typically used to select Forward, Neutral and Reverse, and for this a resistive network is required. REV reads voltages via one of the analog inputs to decide which gear the operator has chosen.

It is important that no other loads are added to this circuit. Such loads can modify the voltage that is read by the controller and thereby cause unreliable gear changes.

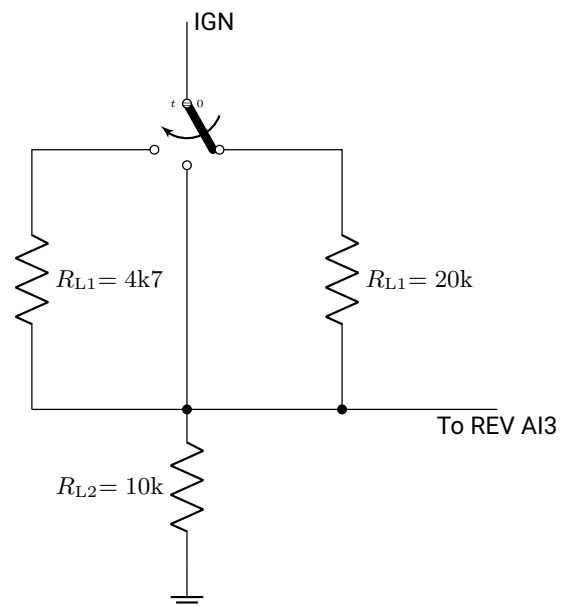


Figure 6: Gear switch circuit



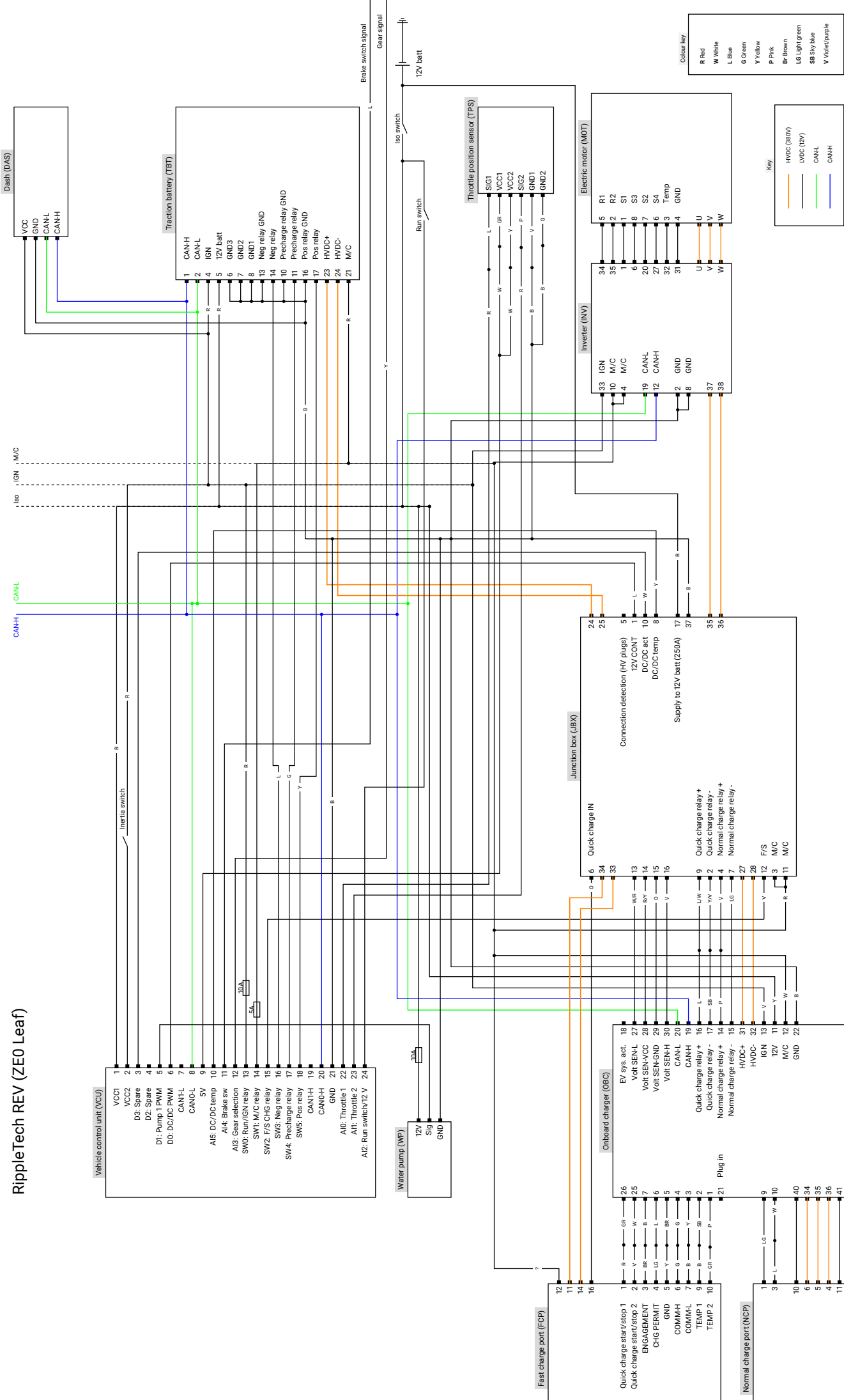
6.2 Connecting REV to a Nissan Leaf ZEO (2009-2012) drive train

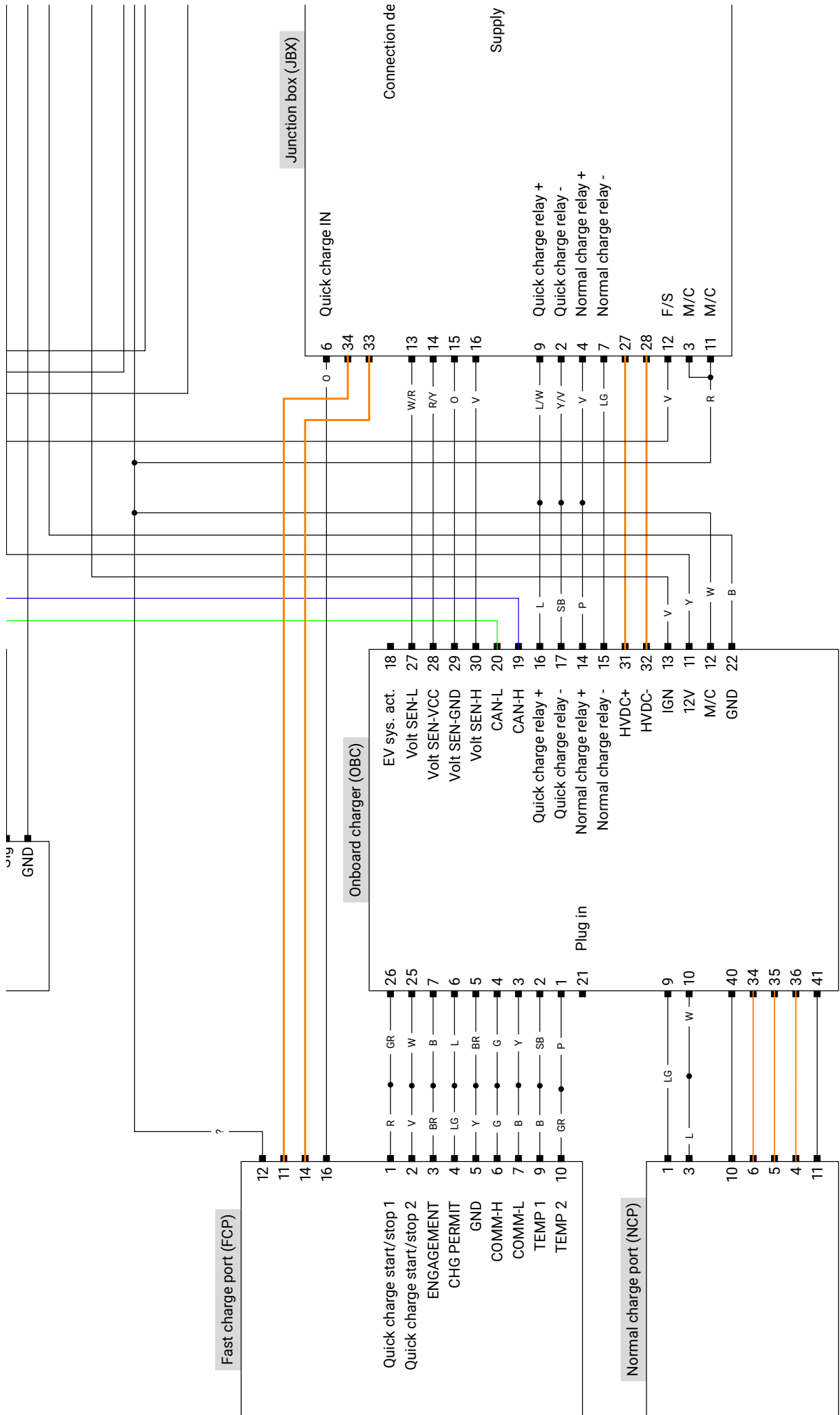
The following instructions are for use with the Nissan Leaf Gen I firmware.

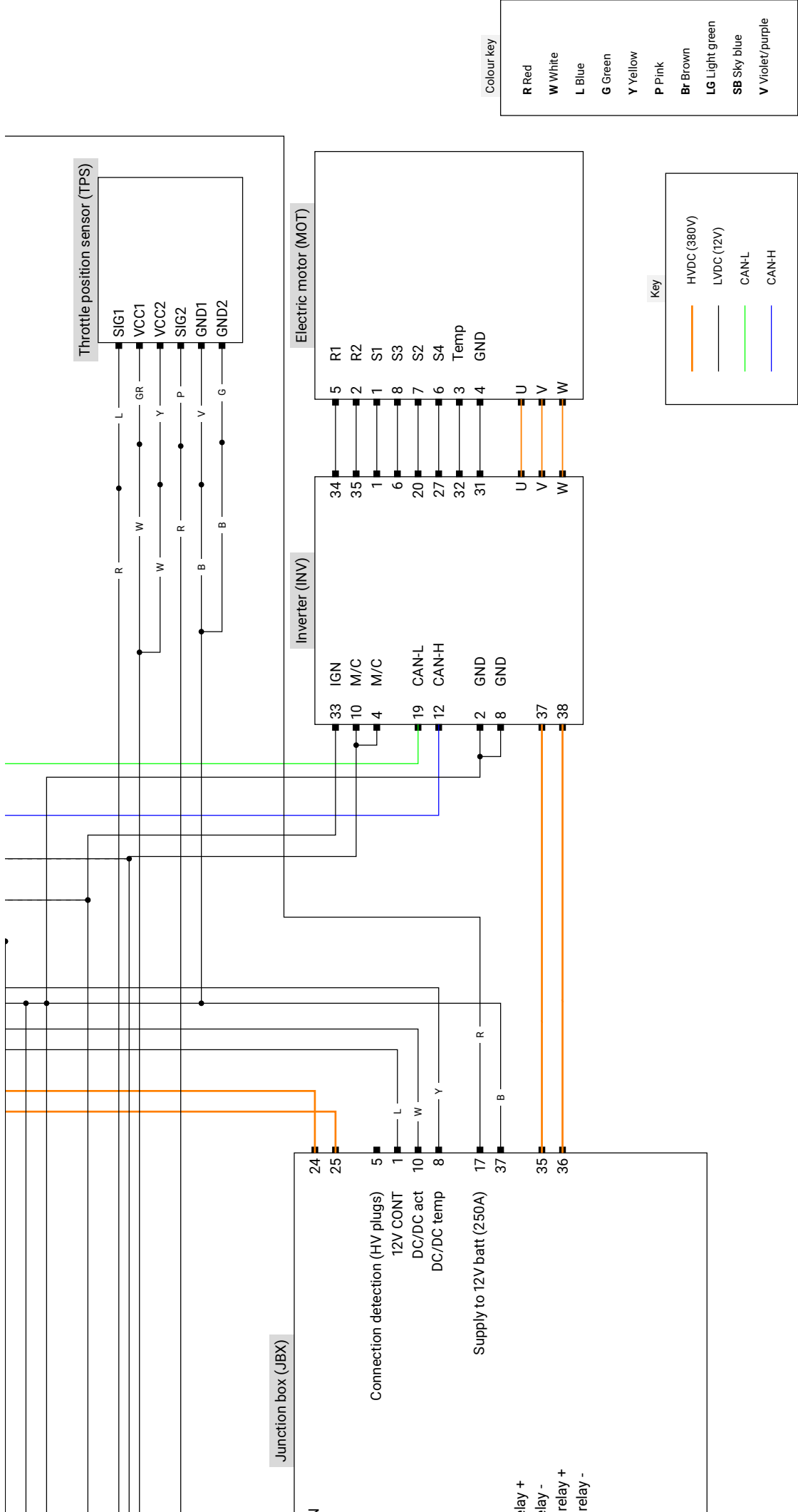
6.2.1 Wiring diagrams

The following page shows the EV system on one page, and the following four pages split the single drawing into four parts (so you can tape them together).

RippleTech REV (ZEO Leaf)

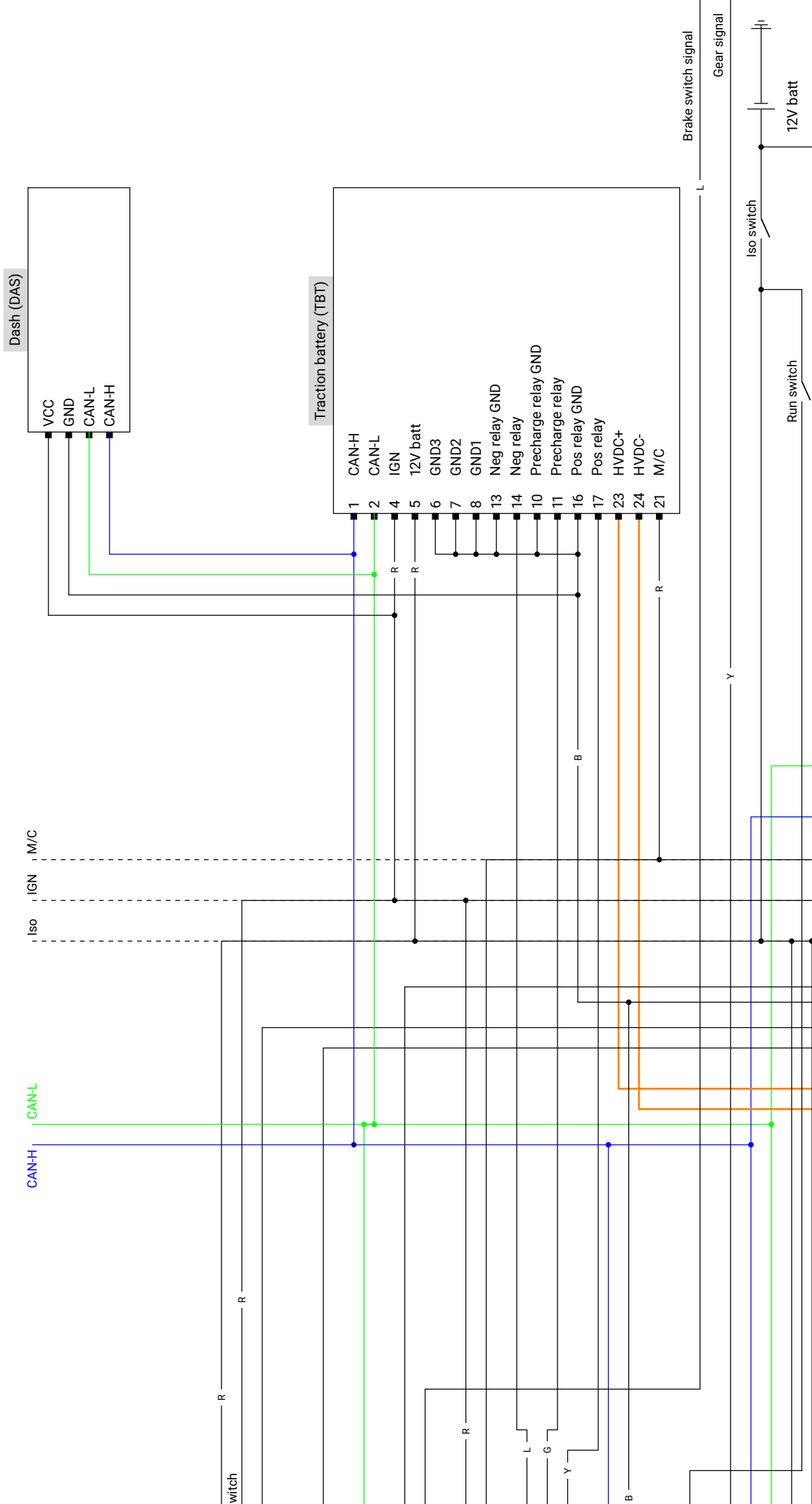






RippleTech REV (ZE0 Leaf)







6.2.2 Connectors

The pins on the following connectors match the pins on the system diagram. The high voltage pins are self evident. Pin 17 on the Junction box is an M8 stud.

Connector No.	F11
Connector Name	DC/DC-J/B
Connector Type	RH12FB

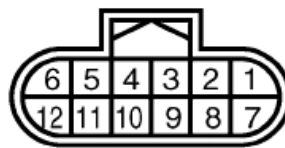


Figure 7: Connector F11 (Junction box)

Connector No.	F12
Connector Name	DC/DC-J/B
Connector Type	RH04FB

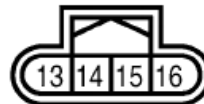


Figure 8: Connector F12 (Junction box)

6.3 Connecting REV to a Nissan Leaf AZE0 (2013-2018) drive train

6.3.1 Wiring diagrams

The following page shows the EV system on one page, and the following four pages split the single drawing into four parts (so you can tape them together).



Connector No.	B24
Connector Name	LI-ION BATTERY
Connector Type	Yazaki_7283-8750-30



Figure 9: Connector B24 (Traction battery)

Connector No.	B26
Connector Name	ON BOARD CHARGER
Connector Type	RH12FB

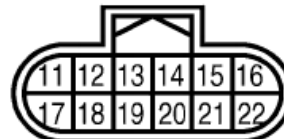


Figure 10: Connector B26 (Onboard charger)

Connector No.	B27
Connector Name	ON BOARD CHARGER
Connector Type	RH08FB



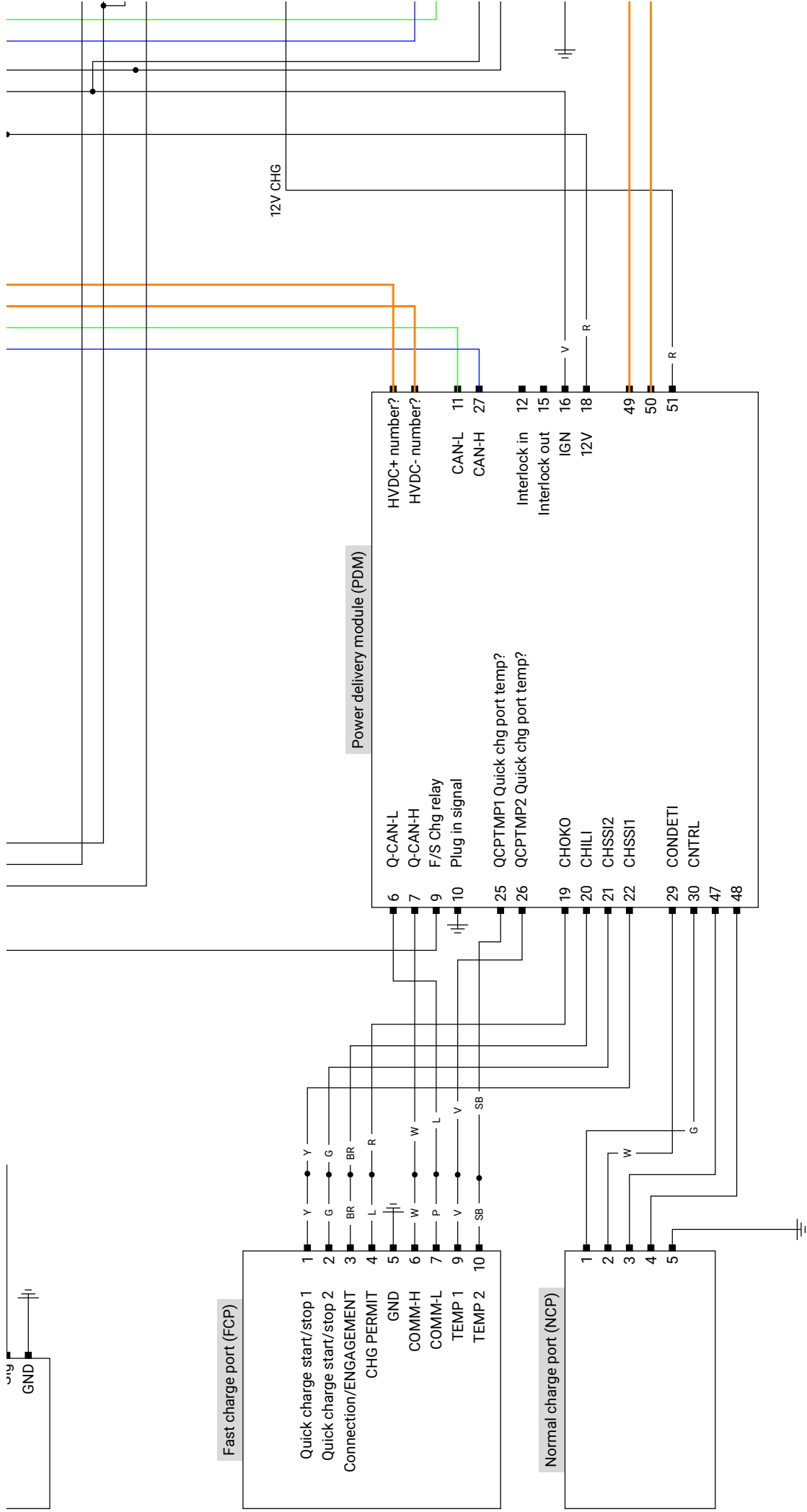
Figure 11: Connector B27 (Onboard charger)

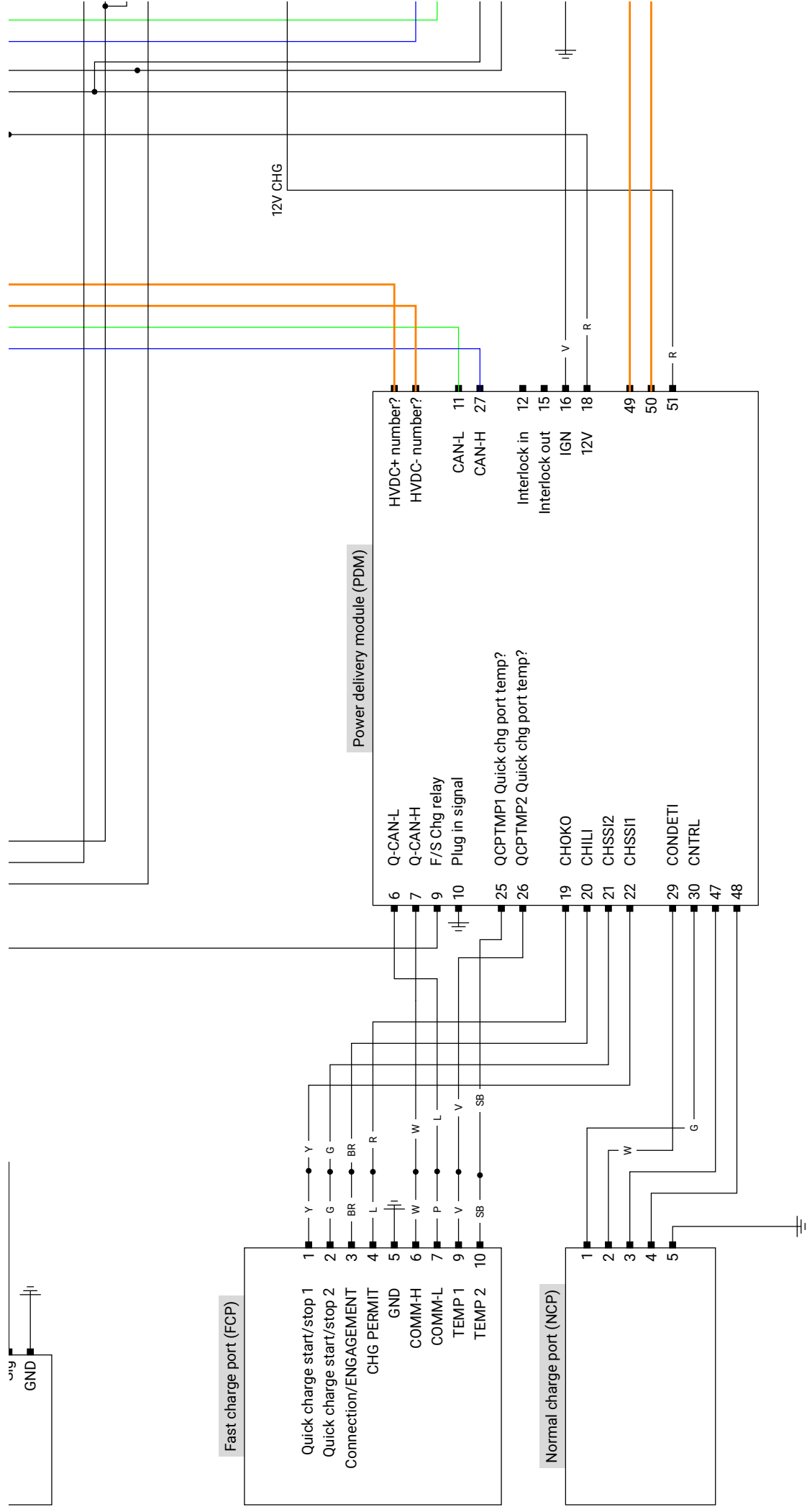


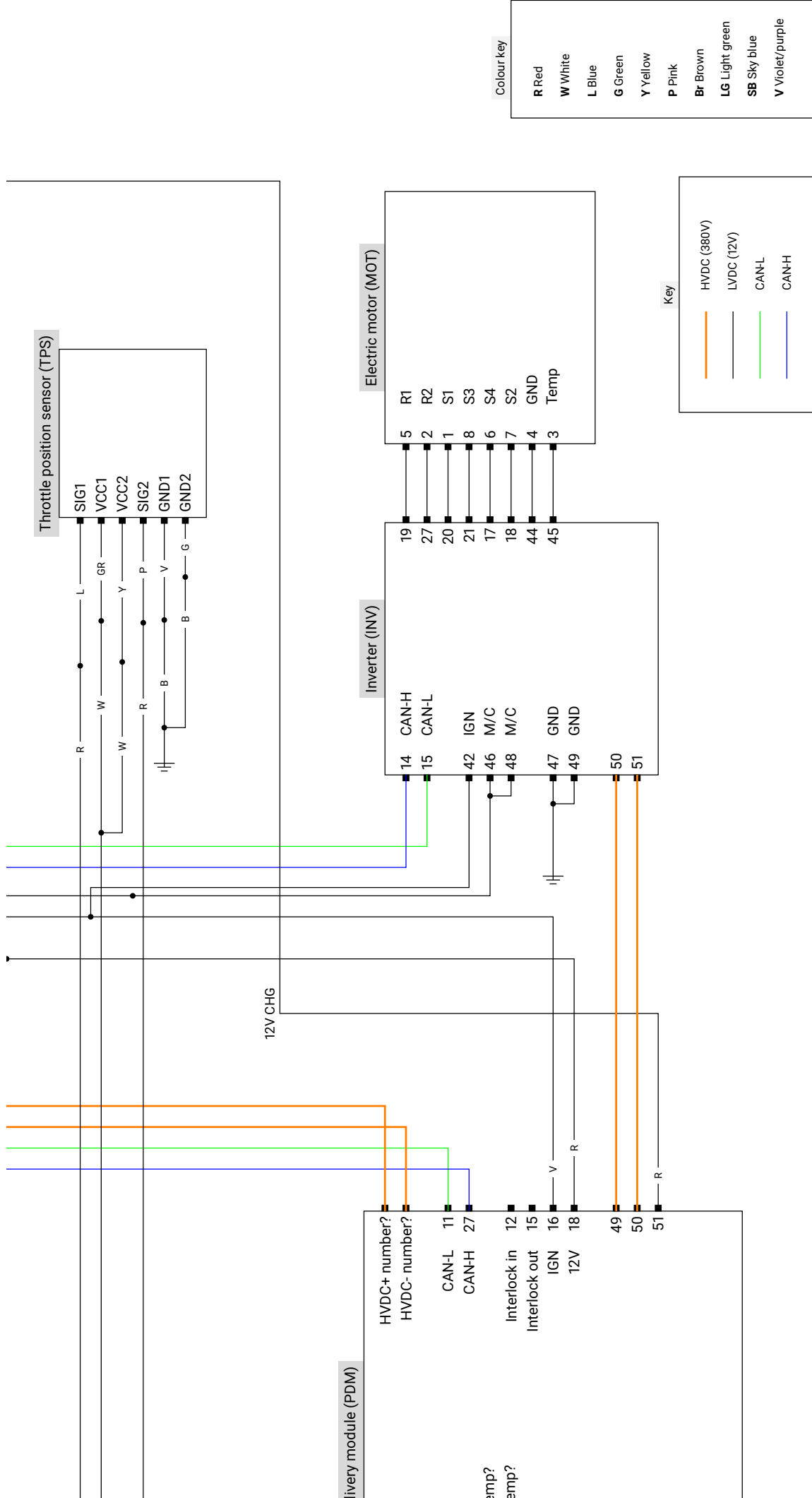
Connector No.	E67
Connector Name	ELECTRIC WATER PUMP 1
Connector Type	RS04FG



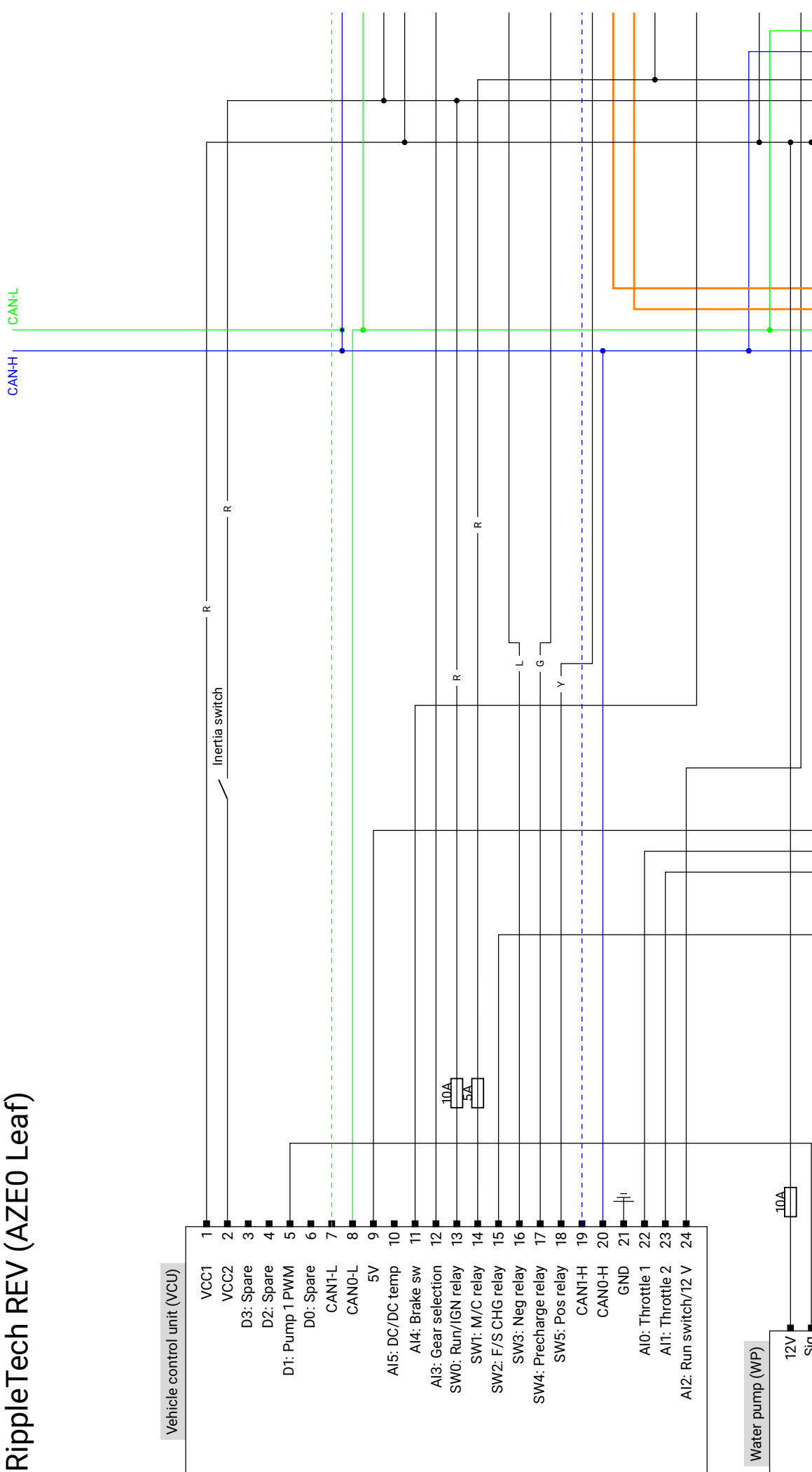
Figure 12: Connector E67 (Water pump)







RippleTech REV (AZE0 Leaf)



6.3.2 Connectors

6.4 Connecting REV to a Nissan Leaf ZE1 (2018+) drive train

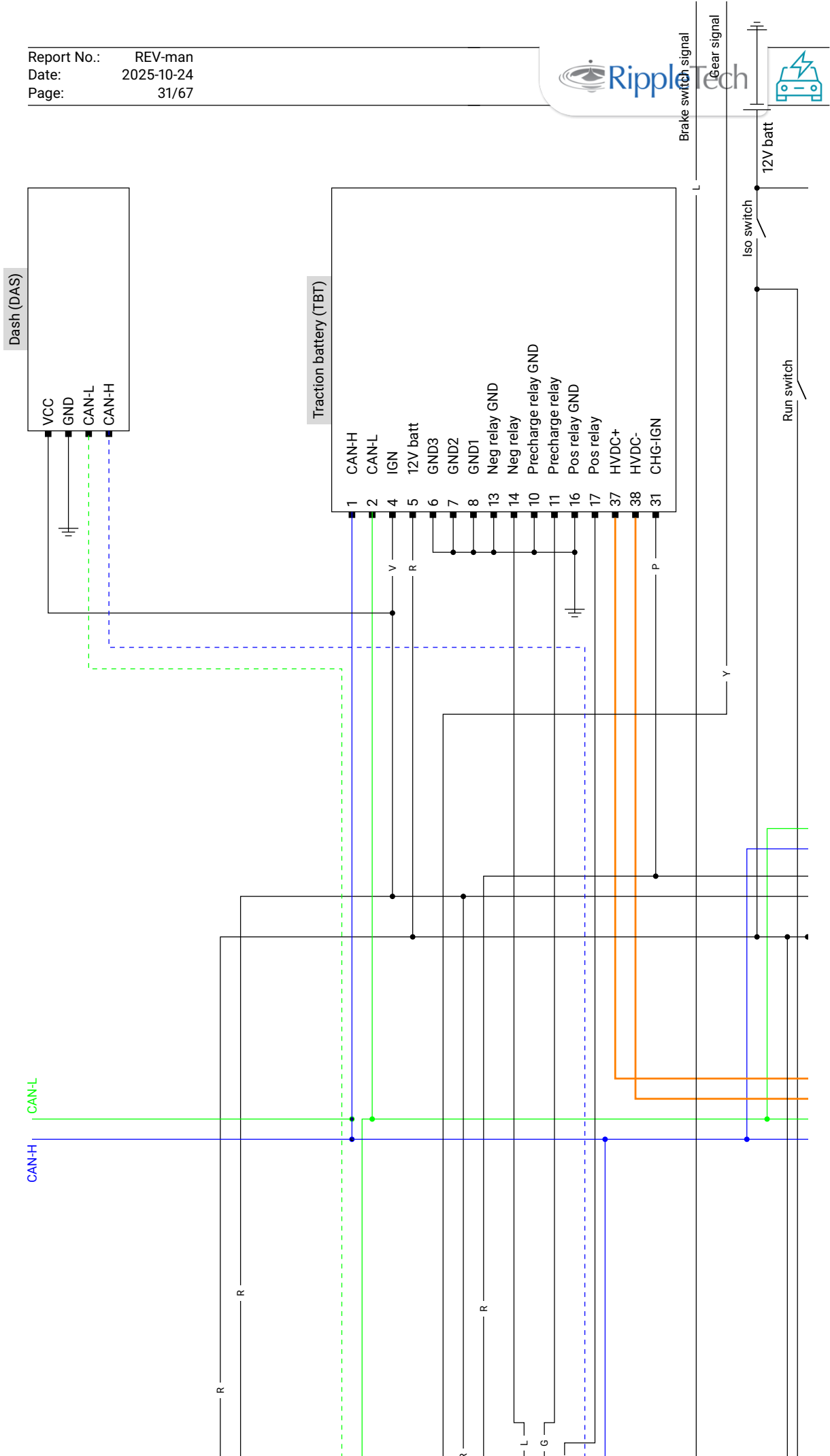
The ZE1 drive train is essentially the same as the AZE0 except for the inverter which has higher output. Note to utilise the higher power from standard 80kw then adjust the settings in the wifi web settings page.

ZE1 inverters (2018-2026) |291A0-5SA1A - 110kW inverter| |291A0-5SA1B - 110kW inverter| |291A0-5SA0A - 110kW inverter| |291A0-5SN0A - 160kW inverter| |291A0-5SN1A - 160kW inverter|

If buying parts you can use an AZE0 drive train and find a ZE1 inverter at a later date and swap out if a higher performance is needed.

6.4.1 Wiring diagrams

The following page shows the EV system on one page, and the following four pages split the single drawing into four parts (so you can tape them together).





Connector No.	H17
Connector Name	PDM (POWER DELIVERY MODULE)
Connector Color	–



Terminal No.	Color of Wire	Signal Name
49	–	–
50	–	–

Figure 13: Connector H17 (Power Delivery Module)



Connector No.	F32
Connector Name	PDM (POWER DELIVERY MODULE)
Connector Color	–

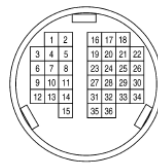


Terminal No.	Color of Wire	Signal Name
51	B/Y	–

Figure 14: Connector F32 (Power Delivery Module)



Connector No.	F23
Connector Name	PDM (POWER DELIVERY MODULE)
Connector Color	GREEN



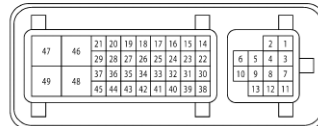
Terminal No.	Color of Wire	Signal Name
1	—	—
2	—	—
3	—	—
4	—	—
5	—	—
6	L	Q-CAN-L
7	W	Q-CAN-H
8	—	—
9	GR	QCRLV
10	LG	CSTATE
11	G	EV CAN-L
12	O	INTERLOCK_IN
13	—	—
14	—	—
15	P	INTERLOCK_OUT
16	V	IGN
17	—	—
18	R	BAT
19	R	CHOKO
20	BR	CHILI

Terminal No.	Color of Wire	Signal Name
21	G	CHSSI2
22	Y	CHSSI1
23	—	—
24	—	—
25	SB	QCPTMP2
26	V	QCPTMP1
27	L	EV CAN-H
28	—	—
29	W	CONDET1
30	BR	CNTRL
31	—	—
32	—	—
33	—	—
34	—	—
35	—	—
36	—	—

Figure 15: Connector F23 (Power Delivery Module)



Connector No.	F13
Connector Name	TRACTION MOTOR INVERTER
Connector Color	BLACK



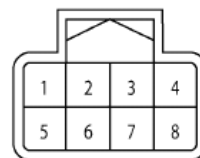
Terminal No.	Color of Wire	Signal Name
1	–	–
2	–	–
3	–	–
4	–	–
5	–	–
6	–	–
7	–	–
8	–	–
9	–	–
10	–	–
11	–	–
12	–	–
13	–	–
14	L	EV SYSTEM CAN-H
15	G	EV SYSTEM CAN-L
16	–	–
17	P	REZ_S2
18	L	REZ_S4
19	R	REZ_R1
20	B	REZ_S1

Terminal No.	Color of Wire	Signal Name
21	W	REZ_S3
22	–	–
23	–	–
24	–	–
25	–	–
26	–	–
27	G	REZ_R2
28	–	–
29	–	–
30	–	–
31	–	–
32	–	–
33	–	–
34	–	–
35	–	–
36	–	–
37	–	–
38	–	–
39	–	–
40	–	–
41	–	–
42	LG	IGN_SW
43	–	–
44	O	TMGND
45	Y	TM
46	G	VB1
47	B	VBGND1
48	G	VB2
49	B	VBGND2

Figure 16: Connector F13 (Traction Motor Inverter)



Connector No.	F 14
Connector Name	TRACTION MOTOR
Connector Color	BLACK



Terminal No.	Color of Wire	Signal Name
1	B	–
2	G	–
3	Y	–
4	O	–
5	R	–
6	P	–
7	L	–
8	W	–

Figure 17: Connector F14 (Traction Motor)



Connector No.	H3
Connector Name	LI-ION BATTERY
Connector Color	—



Terminal No.	Color of Wire	Signal Name
37	O	—
38	O	—
39	SHIELD	—

Figure 18: Connector H3 (Traction Battery)

Connector No.	B24
Connector Name	LI-ION BATTERY
Connector Color	GREEN




Terminal No.	Color of Wire	Signal Name
1	L	EV CAN-H
2	G	EV CAN-L
3	—	—
4	V	IGN

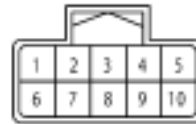
Terminal No.	Color of Wire	Signal Name
5	R	BAT
6	B	GND3
7	B	GND2
8	B	GND1
9	—	—
10	B	PRE CHG GND
11	G	PRE CHG V
12	—	—
13	B	RLY N GND
14	L	RLY N V
15	—	—
16	B	RLY P GND
17	Y	RLY P V
18	—	—
19	—	—
20	—	—

Terminal No.	Color of Wire	Signal Name
21	—	—
22	—	—
23	—	—
24	—	—
25	—	—
26	—	—
27	—	—
28	—	—
29	—	—
30	—	—
31	P	CHG IGN
32	—	—
33	—	—
34	—	—
35	—	—
36	—	—

Figure 19: Connector B24 (Traction Battery)



Connector No.	E7
Connector Name	QUICK CHARGE PORT
Connector Color	BLACK



Terminal No.	Color of Wire	Signal Name
1	Y	–
2	G	–
3	GR	–
4	L	–
5	B	–
6	W	–
7	L	–
8	–	–
9	V	–
10	S B	–

Figure 20: Connector E7 (Quick Charge Port)

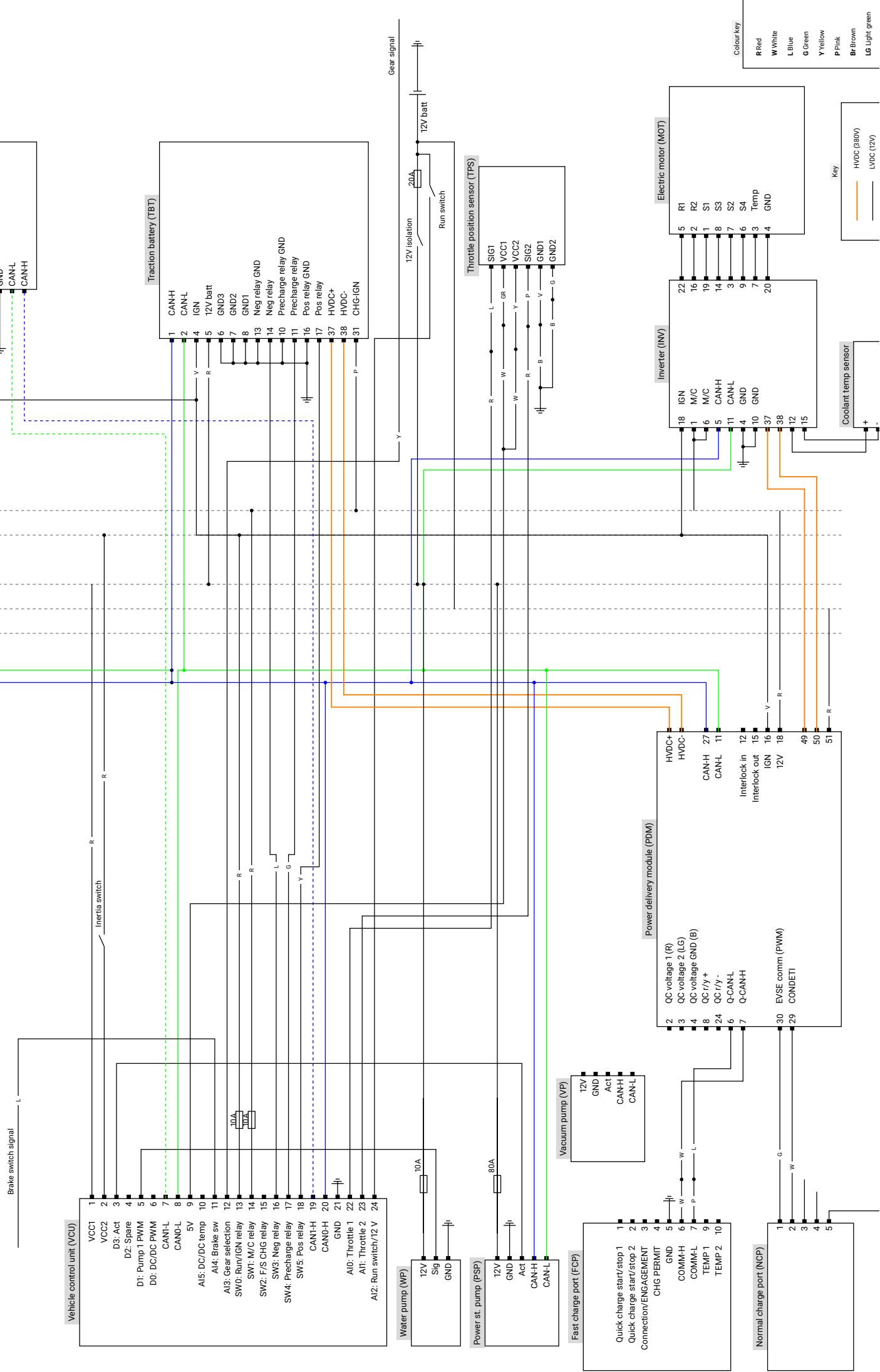


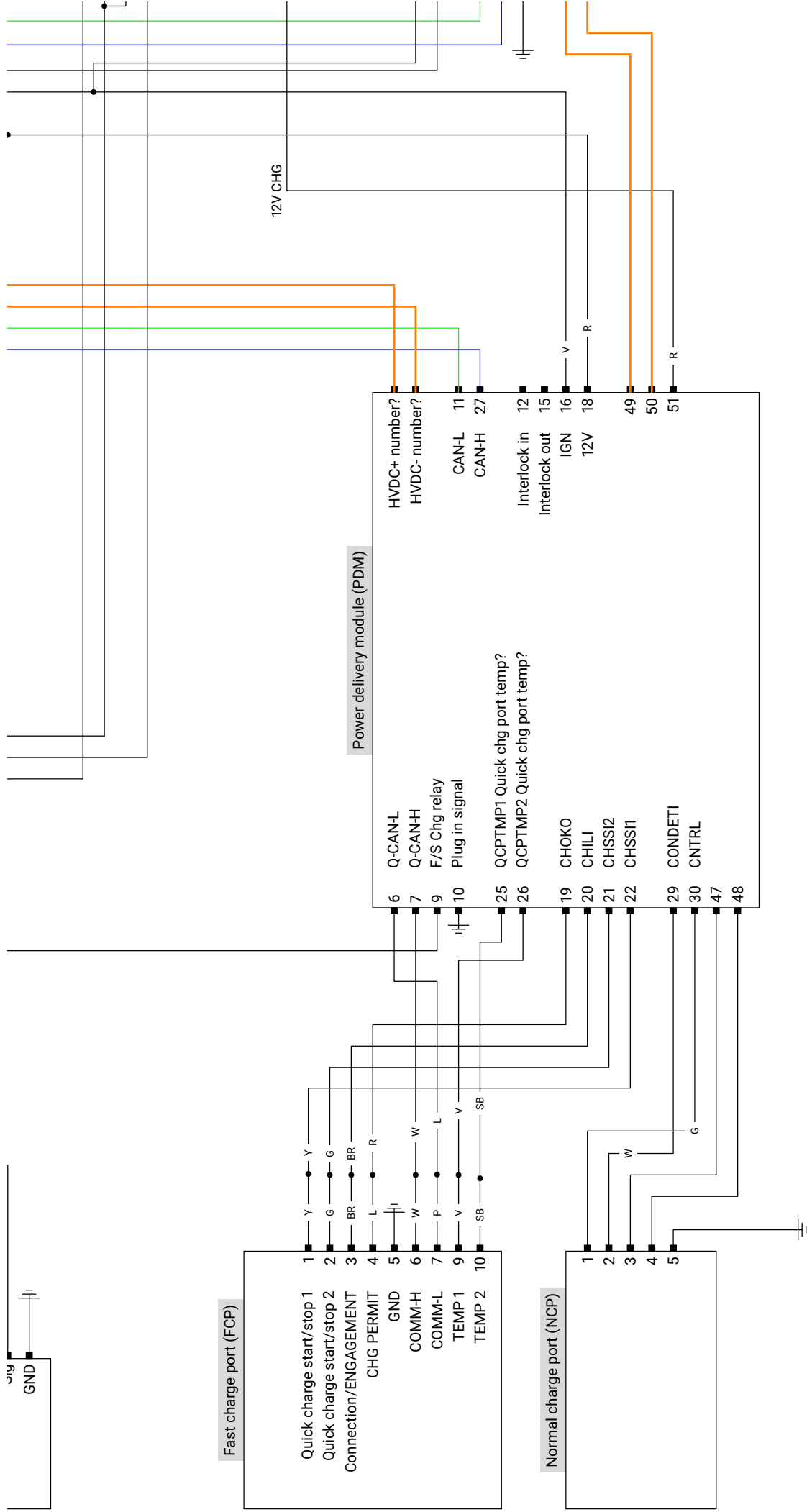
Connector No.	E 25
Connector Name	NORMAL CHARGE PORT
Connector Color	BLACK

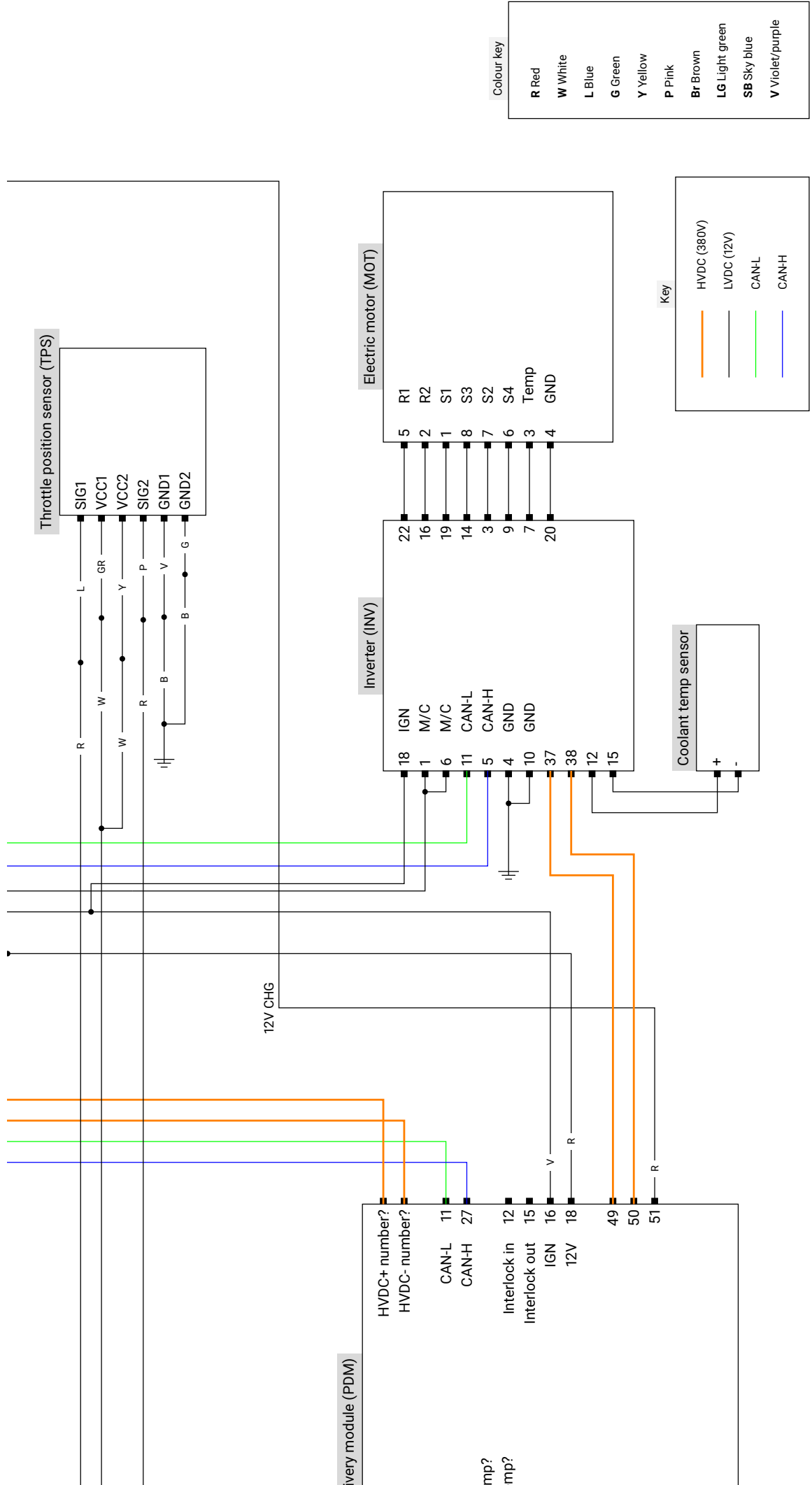


Terminal No.	Color of Wire	Signal Name
1	LG	–
2	W	–

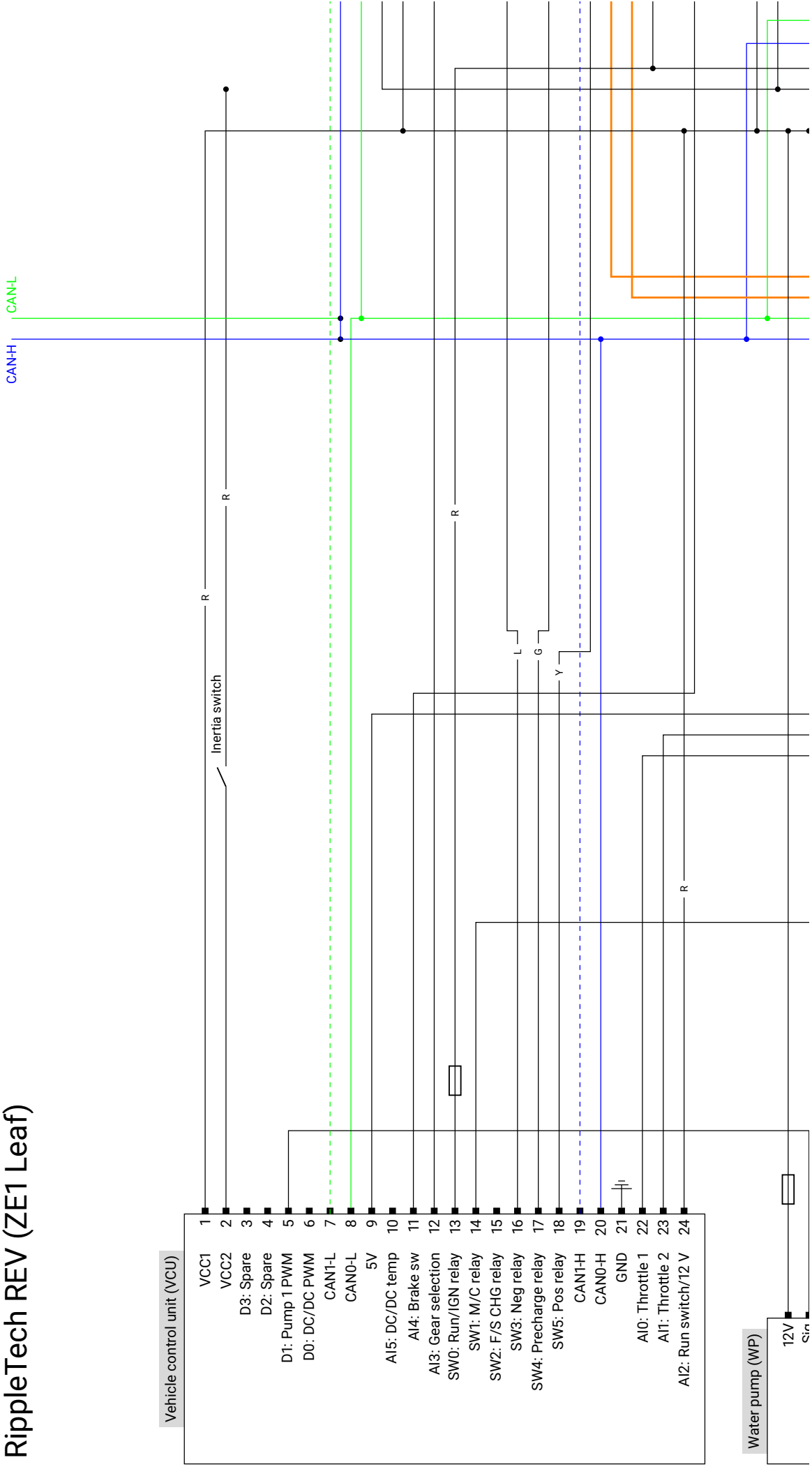
Figure 21: Connector E25 (Normal Charge Port)



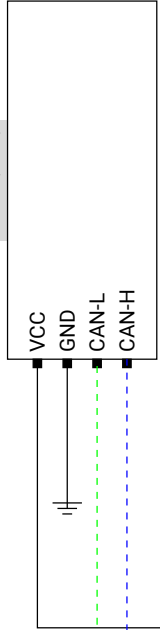




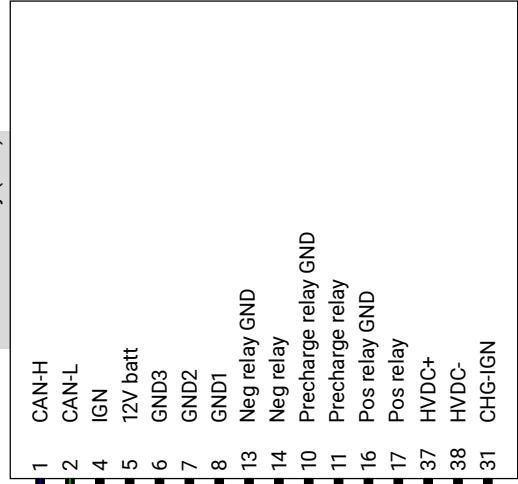
RippleTech REV (ZE1 Leaf)



Dash (DAS)

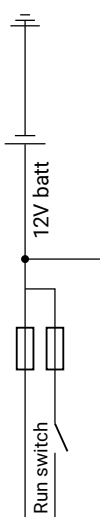


Traction battery (TBT)

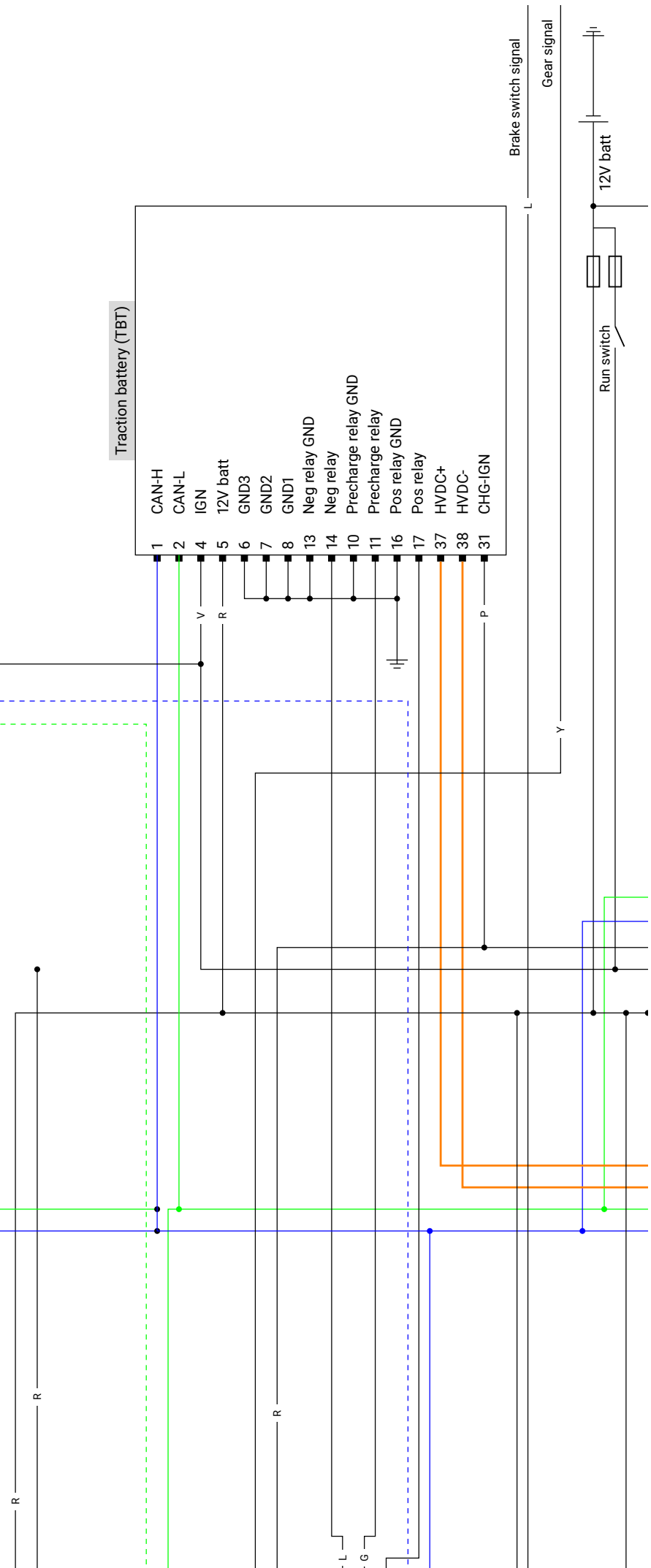


Brake switch signal

Gear signal



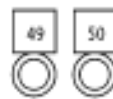
CAN-H
CAN-L





6.4.2 Connectors

Connector No.	H17
Connector Name	PDM (POWER DELIVERY MODULE)
Connector Color	–



Terminal No.	Color of Wire	Signal Name
49	–	–
50	–	–

Figure 22: Connector H17 (Power Delivery Module)

6.5 Nissan Leaf water pumps

These are 4-wire variable speed pumps. The pins, as seen looking at face of plug, are:

1 4
2 3

The numbers are also written on the socket of the pump. The pins are as follows (with colours found on Nissan looms):

Pin	Colour	Colour	Colour	Function
1	Br	L	Br	Vcc (12v)
2	W	K	B/Y	Gnd
3	LG	R	G	Activation signal (PWM into pump)
4	Y	Y	W	Signal (PWM out of pump)



Connector No.	F32
Connector Name	PDM (POWER DELIVERY MODULE)
Connector Color	–



Terminal No.	Color of Wire	Signal Name
51	B/Y	–

Figure 23: Connector F32 (Power Delivery Module)

6.6 How to install the RippleTech REV Controller.

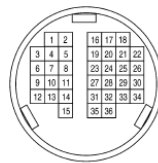
Read the full instructions first for an overview.

Alert: Complete acceleration calibration before first use or whenever any changes are done to the accelerator inputs. Depending on the default calibration and the device, the REV controller may read the accelerator pedal as not at rest and therefore drive forward unexpectedly when you first put into gear. You cannot switch into gear unless the vehicle is stationary and the brake is pressed and the accelerator is not depressed. This is by design. If the vehicle does not go into gear and you have checked the brake signal, check the accelerator is correctly calibrated.

Alert: Complete acceleration calibration before first use or whenever any changes are done to the accelerator inputs. Depending on the default calibration and the device, the REV controller may read the accelerator pedal as not at rest and therefore drive forward unexpectedly when you first put into gear. You cannot switch into gear unless the vehicle is stationary and the brake is pressed and the accelerator is not depressed. This is by design. If the vehicle does not go into gear and you have checked the brake signal, check the accelerator is correctly calibrated.



Connector No.	F23
Connector Name	PDM (POWER DELIVERY MODULE)
Connector Color	GREEN



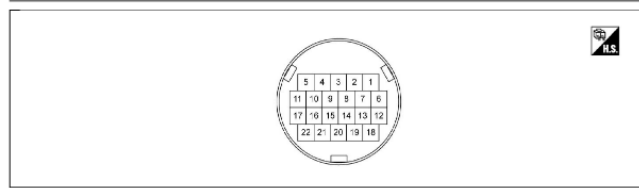
Terminal No.	Color of Wire	Signal Name
1	—	—
2	—	—
3	—	—
4	—	—
5	—	—
6	L	Q-CAN-L
7	W	Q-CAN-H
8	—	—
9	GR	QCRLV
10	LG	CSTATE
11	G	EV CAN-L
12	O	INTERLOCK_IN
13	—	—
14	—	—
15	P	INTERLOCK_OUT
16	V	IGN
17	—	—
18	R	BAT
19	R	CHOKO
20	BR	CHILI

Terminal No.	Color of Wire	Signal Name
21	G	CHSSI2
22	Y	CHSSI1
23	—	—
24	—	—
25	SB	QCPTMP2
26	V	QCPTMP1
27	L	EV CAN-H
28	—	—
29	W	CONDET1
30	BR	CNTRL
31	—	—
32	—	—
33	—	—
34	—	—
35	—	—
36	—	—

Figure 24: Connector F23 (Power Delivery Module)



Nissan Leaf ZE1 Inverter Pinout Connections



-01-000145925

Connection Descriptions

CAUTION:

- The pinout diagram shown is viewed from the wire side of the inverter connector when attached. The installed vertical orientation of the connector is not as shown in the diagram. Use the bottom notch shown for reference.

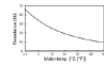
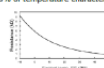
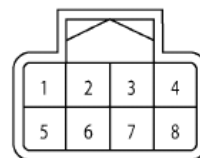
Terminal No. (Color)	Description		Condition	Value (Approx.)
+	Signal name	Input/ Output		
1 (G)	Ground	12V battery power supply	Input	Power switch ON
3 (L)	9 (P)	Traction motor resolver signal (S2 - S4)	Input	Power switch OFF
4 (B)	Ground	GND	—	Always
5 (L)	—	EV system CAN-H	Input/Output	—
6 (G)	Ground	12V battery power supply	Input	Power switch ON
7 (V)	20 (V)	Traction motor temperature sensor	Input	Power switch OFF
				Within ± 50% of temperature characteristics diagram  -03-000165492
10 (B)	Ground	GND	—	Always
11 (G)	—	EV system CAN-L	Input/Output	—
12 (V)	15 (BR)	Coolant temperature sensor	Input	Power switch OFF
				Within ± 50% of temperature characteristics diagram  -02-000165491
18 (LG)	Ground	Power ON power supply	Input	Power switch ON
				Power switch OFF
19 (B)	14 (W)	Traction motor resolver signal (S1 - S3)	Input	Power switch OFF
22 (R)	16 (G)	Traction motor resolver signal (R1 - R2)	Output	Power switch OFF
				8 - 15 Ω

Figure 25: Connector F13a (Traction Motor Inverter)



Connector No.	F 14
Connector Name	TRACTION MOTOR
Connector Color	BLACK



Terminal No.	Color of Wire	Signal Name
1	B	–
2	G	–
3	Y	–
4	O	–
5	R	–
6	P	–
7	L	–
8	W	–

Figure 26: Connector F14 (Traction Motor)



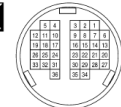
Connector No.	H3
Connector Name	LI-ION BATTERY
Connector Color	—



Terminal No.	Color of Wire	Signal Name
37	O	—
38	O	—
39	SHIELD	—

Figure 27: Connector H3 (Traction Battery)

Connector No.	B04
Connector Name	LI-ION BATTERY
Connector Color	GREEN

Terminal No.	Color of Wire	Signal Name
1	L	EV CAN-H
2	G	EV CAN-L
3	—	—
4	V	IGN

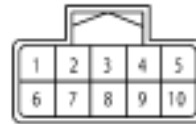
Terminal No.	Color of Wire	Signal Name
5	R	BAT
6	B	GND3
7	B	GND2
8	B	GND1
9	—	—
10	B	PRE CHG GND
11	G	PRE CHG V
12	—	—
13	B	RLY N GND
14	L	RLY N V
15	—	—
16	B	RLY P GND
17	Y	RLY P V
18	—	—
19	—	—
20	—	—

Terminal No.	Color of Wire	Signal Name
21	—	—
22	—	—
23	—	—
24	—	—
25	—	—
26	—	—
27	—	—
28	—	—
29	—	—
30	—	—
31	P	CHG IGN
32	—	—
33	—	—
34	—	—
35	—	—
36	—	—

Figure 28: Connector B24 (Traction Battery)



Connector No.	E7
Connector Name	QUICK CHARGE PORT
Connector Color	BLACK



Terminal No.	Color of Wire	Signal Name
1	Y	–
2	G	–
3	GR	–
4	L	–
5	B	–
6	W	–
7	L	–
8	–	–
9	V	–
10	S B	–

Figure 29: Connector E7 (Quick Charge Port)



Connector No.	E25
Connector Name	NORMAL CHARGE PORT
Connector Color	BLACK



Terminal No.	Color of Wire	Signal Name
1	LG	-
2	W	-

Figure 30: Connector E25 (Normal Charge Port)

6.6.1 Cabling

NB We have found some Nissan Leaf's manufactured have slightly differing colours to the service manual. Therefore we suggest confirming the pins on each plug and corresponding colour wires. All plugs are viewed from the back where the wires are.

There are a fair few wires to connect up and we therefore recommend using temporary connectors on the bench prior to installing into its final location. This allows double checking and fault finding should that be required. We suggest printing out key parts and creating a list to help with what goes where.

Once you have it running you can remove and wire up permanently as a tidy loom with correct wire length in its final destination. We find that PT2.5 push-in Spring connectors on a DIN rail works well.

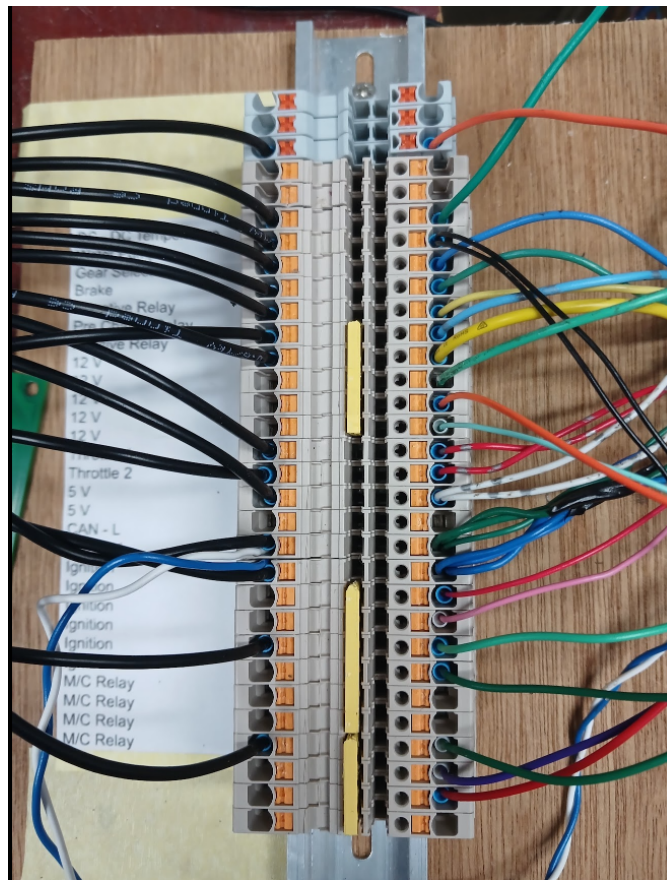


Figure 31: Temporary wire connectors (initial testing)



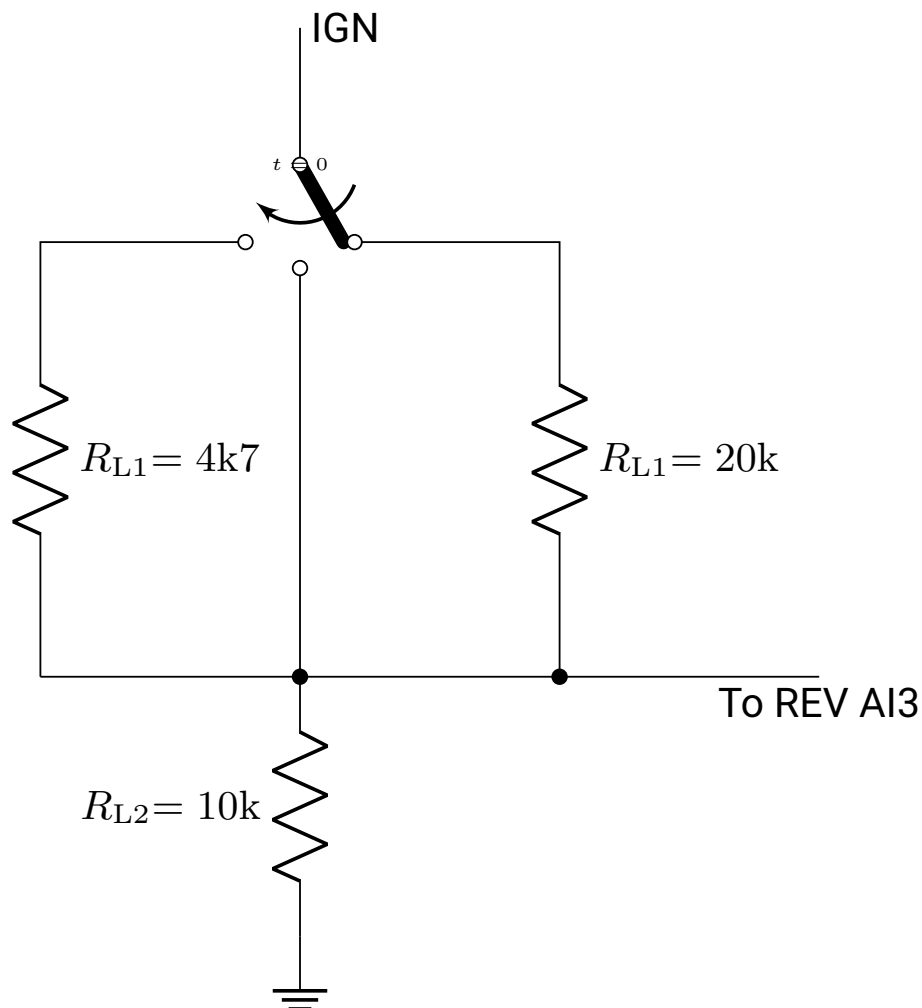
6.6.2 Can Bus cabling

To minimise the noise from other sources that can affect the CAN bus communication, two wires that carry opposite signals are twisted together. This technique ensures that both wires have the same noise level and resistance. You need to check your layout and remove the termination jumpers inside the REV controller, because they can cause problems if there are other terminations on the bus. We usually connect the REV controller to the same place where the inverter and BMS are connected, and we take out the terminations inside the controller. Remember that the CAN bus only needs termination at two ends. You can measure the resistance of each termination with a multimeter. It should be 120 Ohms for each termination, and 60 Ohms for the whole bus. For further information feel free to search on the web introduction to can bus wiring.

6.6.3 GND Connections

You should connect all the GND wires to one common point, and not to various different parts of the vehicle chassis.

1. Gear selector switch diagram. A three way switch is typically used to select Forward, Neutral and Reverse, and for this a resistive network is required. REV reads voltages via one of the analog inputs to decide which gear the operator has chosen.



6.6.4 First run steps:

Confirm the system starts by going through a precharge cycle. A precharge cycle for an EV is a process of gradually charging the high-voltage capacitors in the electric drive system before the main contactors are closed. This is done to avoid damaging in-rush currents that can occur when the capacitors are suddenly connected to the battery pack. A precharge cycle typically takes about 1 second. You should hear multiple clunks from the solenoids in the traction battery (one when the negative and precharge relays close, and another when the precharge relay opens and the positive relay closes). There are various reasons for the precharge to fail. In this case the REV controller will throw a trouble code.

1. Accelerator pedal calibration Before putting into gear, please calibrate the accelerator setting through the web interface. The REV Wi-Fi is only acces-



sible after a power cycle and before going into gear. Once it is in gear, the Wi-Fi is switched off by design.

If you do not get typical values that deviate from off to full throttle eg between 0 and 4096 then check accelerator pedal wiring.

Gear select circuit check Check Gear select voltages of REV = ~6V N= 0V FWD = ~4V

Brake circuit check Check Brake voltages of Brake ON= ~12 and Brake Off = ~0

Accelerator pedal (Nissan Leaf) AI0 & AI1 ~0.8V at rest - ~4.4V at full throttle ~0.4V at rest - ~2.2V at full throttle

6.6.5 Running the EM61 / EM57 motor in reverse

The Nissan Leaf motor is limited to 3000 RPM in reverse. If the motor is mounted such that running "backwards" is required for normal (forward) operation, the following steps must be taken.

1. Swap the "sine" wires in the harness connecting the motor resolver to the inverter.
2. Swap phases V and W between the motor and the inverter.
3. Example below of the EM61

6.6.6 To install the OBDII female connector

This allows Leaf spy to access the traction battery and used for Android instrument solutions using bluetooth OBDII dongle. For Leaf Spy remember to change the settings in leaf spy under "Model Year" and select BMS only etc for the traction battery, or MTR/INV for the inverter.

- Pin 6 = CAN- H
- Pin 14 = CAN-L
- Pin 16 = 12v Power on
- Pin 4, 5 GND

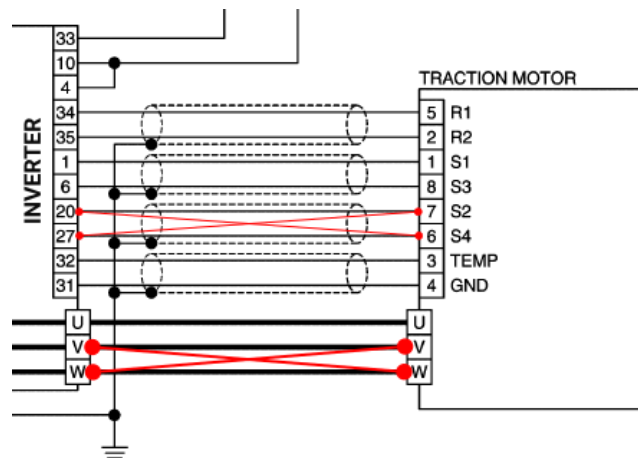


Figure 32: Motor phase and resolver modifications for "backwards" running (red markup)

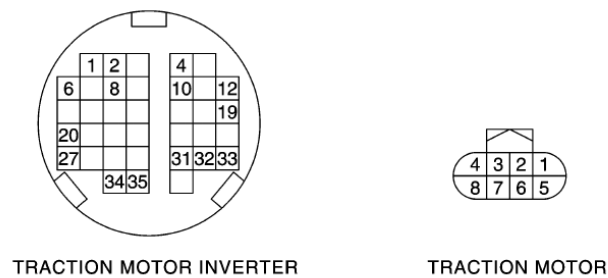


Figure 33: Resolver harness connectors



Figure 34: Connector OBD-II (Diagnostic connector and instrument connectivity)



7 CLIENT DOCUMENTATION

Client documentation@{Client documentation}} % %

7.1 CAN bus messages that are meant for dash displays

Dash status message Sent by dash software every 10-\100 ms. CAN ID is set by CAN_ID_DASH_STATUS.

Byte	7	6	5	4	3	2	1	0
0	<-	-	Status	->	Sp	Sp	Sp	Sp
1	Sp	Sp	Sp	Sp	Sp	Sp	crConReModeReq	
2	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
3	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
4	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
5	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
6	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
7	Sp	Sp	Sp	Sp	Sp	Sp	<-	Ctr ->

Ctr is 0,1,2,3,0,1,2,...

No checksum or CRC.

{ Dash message 1} Sent by VCU every 100 ms. CAN ID is set by CAN_ID_VCU_DASHMESG

Byte	7	6	5	4	3	2	1	0
0	<-	-	-	-	-	HV	voltage	-
1	-	-	->	<-	-	-	-	-
2	-	-	HV	current	-	->	<-	-
3	-	-	Motor	RPM	-	-	-	-
4	-	->	<-	-	-	Throttle pos	-	-
5	->	<-	-	-	-	Speed	-	-
6	-	-	-	->	<-	VCU	state	->
7	Brake	Sp	Sp	Sp	Sp	Sp	<-	Ctr ->

Ctr is 0,1,2,3,0,1,2,...

No checksum or CRC.

The quantities are as follows\+:



Quantity	Unit	Unit sent	Offset
HV batt voltage	V	hV	0
HV current	A	hA	-1024
Motor RPM	RPM	10xRPM	-1024
Throttle pos	%	d%	0
Speed	KPH	KPH	0
VCU state	-	-	0
Brake on	-	-	-

Notes\+:

- Speed is derived from Motor RPM (normally vehicle speed for cars (REV), 0 for boats (REM), and output shaft speed for PTOs (REP)).
- Throttle pos is accelerator pedal for REV, throttle for REM, and speed control knob for REP.
- Brake is always 0 in REP and REM codes.

Dash message 2

Sent by VCU every 100 ms. CAN ID is set by CAN_ID_VCU_DASHMESG

Byte	7	6	5	4	3	2	1	0
0	M/C	Run sw	Neg	Pre	Pos	EVSE in	EVSE on	Alarm
1	<-	Gear	->	B-mode	Limp	Cruise	Sp	Sp
2	<-	-	-	12V	batt	-	-	->
3	<-	-	-	-	-	SoC	-	-
4	-	->	<-	-	-	-	Charge	rate
5	-	-	-	-	->	<-	-	-
6	Brake	vacuum	-	-	->	<-	-	-
7	Gear	switch	voltage	-	->	Sp	<-	Ctr ->

Ctr is 0,1,2,3,0,1,2,...

No checksum or CRC.

The quantities are as follows:



Quantity	Unit	Unit sent	Offset
VCU status	-	-	-128
Gear	-	-	-
Negative relay	-	-	-
Precharge relay	-	-	-
Positive relay	-	-	-
EVSE plugged in	-	-	-
EVSE on	-	-	-
Run switch	-	-	-
MC relay	-	-	-

Notes:

- Gear is always zero for REP

Dash message 3

Sent by VCU every 100 ms. CAN ID is set by CAN_ID_VCU_DASHMESC

Byte	7	6	5	4	3	2	1	0
0	<-	-	-		-	-	-	-
1	-	-	Odo	/hr	meter	-	-	-
2	-	-	-	-	-	-	-	->
3	<-	-	-	Inv	temp	-	-	->
4	<-	-	-	Motor	temp	-	-	->
5	<-	-	Min	batt	temp	-	-	->
6	<-	-	Max	batt	temp	-	-	->
7	Sp	Sp	Sp	Sp	Sp	Sp	<-	Ctr ->

Ctr is 0,1,2,3,0,1,2,...

No checksum or CRC.

The quantities are as follows:



Quantity	Unit	Unit sent	Offset
Odometer / hour meter	km / hr	dkm / dhr	0
Inverter temp	deg	hdeg	0
Motor temp	deg	hdeg	0
Min batt temp	deg	hdeg	-50
Max batt temp	deg	hdeg	-50

Notes:

- Hour meter is used in REM/REP codes, and odometer is used for REV.

Dash message 4

Sent by VCU every 500 ms (or so). CAN ID is set by CAN_ID_VCU_DAS

Byte	7	6	5	4	3	2	1	0
0	<-	-	-		-	-	-	-
1	-	Min	cell	voltage->	<-	-	-	-
2	-	Max	cell	voltage-	-	-	-	-
3	-	->	<-	-	-	-	-	-
4	-	-	-	Capacity remaining	->	<-		
5	-	-	-	Total capacity	-	-		
6	-	-	-	->	Sp	Sp	Sp	Sp
7	Sp	Sp	Sp	Sp	Sp	Sp	<-	Ctr ->

Ctr is 0,1,2,3,0,1,2,...

No checksum or CRC.

Regular DTC message

Sent by VCU every 2000 ms. CAN ID is set by CAN_ID_DTC_BRC

One of the list of warnings and errors is shown at a time.

Byte	7	6	5	4	3	2	1	0
0	<-	-	-	Lamp status-	-	-	->	
1	<-	-	-	DTC type	-	-	->	



Byte	7	6	5	4	3	2	1	0
2	<-	-	-	Code	-	-	-	-
3	-	-	-	-	-	-	-	->
4	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
5	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
6	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp
7	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp

Code is normally RT error/warning code + 256.

7.1.1 OBD-II messages

These use standard ISO 15765-2 single frame messaging, specifically:

Byte	7
0	Data length
1	Mode
2	PID
3	Data
4	Data
5	Data
6	Data
7	Sp

In the messages below the data length is always 0x06 and the mode is always 0x41 (response). The VCU responds with these messages when sent a 0x7DF message with 0x01 as the mode and the PID given below.

Status message (PID 0x02)

Byte	7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	1	0
1	0	1	0	0	0	0	0	1
2	0	0	0	0	0	0	1	0
3	Sp	Sp	vcuMode	brakeOn F2	F	N	R	
4	Run sw	MC	Neg	Pre	Pos	EVSE	EVSEon	Alarm
5	Limp	Sp	Sp	Sp	Sp	Sp	Sp	Sp



Byte	7	6	5	4	3	2	1	0
1	0	1	0	0	0	0	0	1
2	0	0	0	0	0	1	0	0
3	<-	-	-	Speed-	-	-	-	->
4	<-	-	-	-	-	-	-	-
5	-	-	-	Odometer	-	-	-	-
6	-	-	-	-	-	-	-	->
7	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp

Mapping:

- Speed is in kph
- Odometer is in deci-kilometers
- Hour meter is in deci-hours

Notes\+:

- In REP and REM codes the odometer is actually an hour meter.

Drive information (PID 0x05)

Byte	7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	1	0
1	0	1	0	0	0	0	0	1
2	0	0	0	0	0	1	0	1
3	<-	-	-	MotorRPM	-	-	-	-
4	<-	-	-	-	-	-	-	->
5	<-	-	-	Motor temp	-	-	-	->
6	<-	-	-	Inv temp	-	-	-	->
7	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp

Mapping:

- RPM has an offset of 2048.
- Temperatures are in half degrees and have an offset of 50.



Notes\+:

Traction battery information (PID 0x06)

Byte	7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	1	0
1	0	1	0	0	0	0	0	1
2	0	0	0	0	0	1	1	0
3	<-	-	-	SoC	-	-	-	->
4	<-	-	-	Charge rate	-	-	-	->
5	<-	-	-	Min temp	-	-	-	->
6	<-	-	-	Max temp	-	-	-	->
7	Sp	Sp	Sp	Sp	Sp	Sp	Sp	Sp

Mapping\+:

- State of charge is in increments of half a percent.
- Charge rate is in half amps.
- Temperatures are in half degrees and have an offset of 50.

Notes\+:

7.2 Warning codes

Number	Description
0	No warning
1	Dash processor over loaded
2	VCU: Low 12V battery voltage
3	VCU: Low HVDC voltage
4	VCU: High HVDC voltage
5	VCU: High inverter temperature
6	VCU: Low traction battery cell voltage
7	VCU: High traction battery cell voltage
8	VCU: Low traction battery temperature



Number	Description
9	VCU: High traction battery temperature
10	VCU: HV battery maintenance switch disconnected
11	VCU: IR sensor fault
12	VCU: Traction battery isolation warning
13	VCU: High motor temperature
14	VCU: Regen is currently limited
15	VCU: In Limp Mode
16	VCU: Low traction battery state of charge (SOC)
17	VCU: High current
18	VCU: Not possible to go to standby
19	VCU: Not possible to send CAN data on bus 0
20	VCU: Not possible to send CAN data on bus 1
21	VCU: Negative contactor failure
22	VCU: Traction battery temperature too low
23	VCU: 12V batt high voltage limp limit
24	VCU: 12V batt low voltage limp limit

7.3 Error codes

These are all negative.

Number	Description
-0	No error
-1	Helm: Stale VCU CAN data
-2	VCU: Invalid settings!
-3	VCU: Low 12V battery voltage



Number	Description
-4	VCU: Precharge failed
-5	VCU: HV too high at start of precharge
-6	VCU: Positive or precharge contactor failure
-7	VCU: Abnormal throttle reading!
-8	VCU: Error connecting to WiFi
-9	VCU: Error connecting to MQTT
-10	VCU: Stale dash CAN data
-11	VCU: Stale inverter CAN data
-12	VCU: High voltage harness
-13	VCU: State machine error
-14	VCU: HVDC voltage too low for too long.
-15	VCU: HVDC voltage too low
-16	VCU: Traction battery cell voltage too high
-17	VCU: Motor temperature limp limit
-18	VCU: Inverter temperature limp limit
-19	VCU: traction batt low SoC limp limit
-20	VCU: Unable to send CAN data on bus 0
-21	VCU: Unable to send CAN data on bus 1